

14 — Configuration Reference

Audience: whoever edits `.env` or a Helm values file. **Companion docs:** [02 Deployment](#) · [12 VocX](#) · [09 Backup](#)

1. How configuration works

Every Python service uses `pydantic-settings` with its **own env prefix**, subclassing `BaseServiceSettings` from the shared core.

Service	Prefix
register	<code>REGISTER_</code>
access	<code>ACCESS_</code>
gateway	<code>GATEWAY_</code>
atlas	<code>ATLAS_</code>
vocx	<code>VOCX_</code>
stt	<code>STT_</code>
pulse	<code>PULSE_</code>
workflows / orchestrator / notifier	<code>WORKFLOWS_</code>

The compose file supplies each service's variables, and a **small set of top-level `${...}` substitutions** comes from `deploy/compose/.env`. That is the file you edit on a box — not the compose file itself.

`.env` syntax

Compose's parser accepts **both** forms:

```
STT_CPU_THREADS=4
PULSE_SMTP_HOST: smtp.gmail.com
```

Both reach the container. Verify what actually landed rather than reasoning about it:

```
docker compose -f deploy/compose/docker-compose.yml exec pulse env | grep SMTP
```

Container environment is captured at creation

A container reads its environment **when it is created**, not when it starts. After an `.env` change:

```
docker compose -f deploy/compose/docker-compose.yml up -d <service>    # recreates
# NOT: docker compose restart <service>                                # keeps the old env
```

2. The `.env` variables that actually exist

These are the substitutions the compose file reads. Everything else is fixed in the compose file itself.

Secrets — change every one of these in production

Variable	Default (dev)	What it is
<code>PRISM_DB_PASSWORD</code>	<code>prism</code>	PostgreSQL password
<code>MINIO_ROOT_USER</code> / <code>MINIO_ROOT_PASSWORD</code>	<code>prism</code> / <code>prism-secret</code>	Object store credentials
<code>INTERNAL_SIGNING_SECRET</code>	<code>compose-internal-signing-secret</code>	Signs the internal context. Shared by gateway, register, vocx
<code>SVC_ATLAS_KEY</code>	<code>compose-svc-atlas</code>	The <code>svc_atlas</code> principal
<code>SVC_VOX_KEY</code>	<code>compose-svc-vox</code>	The <code>svc_vox</code> principal
<code>SVC_PULSE_KEY</code>	<code>compose-svc-pulse</code>	The <code>svc_pulse</code> principal
<code>SVC_WORKFLOWS_KEY</code>	<code>compose-svc-workflows</code>	The <code>svc_workflows</code> principal
<code>SVC_GATEWAY_KEY</code>	<code>compose-svc-gateway</code>	<code>svc_gateway</code> — carries no authority of its own ; it only ever delegates a user
<code>SVC_ADVAYA_KEY</code>	<code>compose-svc-advaya</code>	The Advaya boundary lane
<code>VOCX_FRONT_KEY</code>	<code>compose-vocx-front</code>	VocX's own front door — only the gateway's injected key is accepted
<code>STT_API_KEY</code>	<code>compose-stt-key</code>	STT front door
<code>PULSE_API_KEYS</code>	<code>compose-pulse-key</code>	PULSE front door
<code>WORKFLOWS_API_KEYS</code>	<code>compose-workflows-key</code>	Orchestrator front door
<code>WORKFLOWS_PAYLOAD_ENCRYPTION_KEY</code>	<i>(empty)</i>	Set in production. AES-256 key, base64url, exactly 32 bytes decoded
<code>ANTHROPIC_API_KEY</code> , <code>WORKFLOWS_ANTHROPIC_API_KEY</code>	<i>(empty)</i>	Empty ⇒ the deterministic stub runs

Edge and posture

Variable	Default	Meaning
<code>EDGE_HTTP_PORT</code> / <code>EDGE_HTTPS_PORT</code>	<code>80</code> / <code>443</code>	Published edge ports
<code>DEV_PORTS_BIND</code>	<code>127.0.0.1</code>	Interface the dev ports bind to. Keep it loopback in production
<code>GATEWAY_REQUIRE_AUTH</code>	<code>false</code>	true in production
<code>GATEWAY_OIDC_ISSUER</code>	(empty)	e.g. <code>http://dex:5556/dex</code> — as reachable <i>inside</i> the network
<code>GATEWAY_OIDC_AUDIENCE</code>	<code>prism</code>	
<code>GATEWAY_CORS_ORIGINS</code>	(empty)	Only needed if the UI is hosted off-origin
<code>GOOGLE_SSO_CLIENT_ID</code>	—	Dex's upstream Google connector
<code>UI_DEX_URL</code> , <code>UI_USE_REAL_API</code>	—	UI build-time wiring

VocX

Variable	Default	Meaning
<code>VOCX_STT_BACKEND</code>	<code>api</code>	<code>api</code> · <code>faster_whisper</code> · <code>stub</code>
<code>VOCX_REDIRECT_URI</code>	<code>localhost</code>	As the browser sees it. Must match Google Console exactly
<code>VOCX_S3_PUBLIC_ENDPOINT_URL</code>	<code>http://localhost:9000</code>	Presigned playback URLs are opened by the browser
<code>VOCX_AUDIO_RETENTION_DAYS</code>	<code>0</code>	<code>0</code> = keep forever
<code>VOCX_DEV_UI</code>	<code>true</code> in dev	The <code>/vocx/v1/dev-ui</code> console. Pinned off by the prod-posture overlay
<code>VOCX_STT_PRIMING</code>	<code>true</code>	Prime Whisper with client names and finance terms
<code>VOCX_EXTRACT_GLOSSARY</code> / <code>_FEW_SHOT</code> / <code>_STRUCTURED</code>	<code>true</code>	Extraction quality switches

STT

Variable	Default	Meaning
<code>STT_CPU_THREADS</code>	<code>0</code>	<code>0</code> ⇒ CTranslate2 reads the host's cores. See §5

PULSE

Variable	Meaning
<code>PULSE_SCHEDULER_ENABLED</code>	Whether PULSE schedules its own scans
<code>PULSE_DISABLE_GDELT</code>	Turn off one source
<code>PULSE_SMTP_HOST</code> / <code>_PORT</code> / <code>_USER</code> / <code>_PASS</code> / <code>_FROM</code> / <code>_FROM_NAME</code>	Digest email

Workflows / orchestrator / notifier

Variable	Meaning
<code>WORKFLOWS_REQUIRE_AUTH</code>	Verify the approver's bearer rather than trusting a <code>by</code> field
<code>WORKFLOWS_APPROVER_NOTIFY</code>	Who is told a decision is awaited
<code>WORKFLOWS_NOTIFICATIONS_ENABLED</code> , <code>_NOTIFY_CHANNELS</code> , <code>_NOTIFY_WEBHOOK_URL</code> , <code>_SMS_WEBHOOK_URL</code> , <code>_OPS_WEBHOOK_URL</code>	Delivery
<code>WORKFLOWS_SMTP_HOST</code> / <code>_PORT</code> / <code>_USERNAME</code> / <code>_PASSWORD</code> / <code>_FROM</code>	Mail
<code>WORKFLOWS_NOTIFIER_INTERVAL_SECONDS</code> , <code>_NOTIFIER_MAX_ATTEMPTS</code>	Outbox drain
<code>WORKFLOWS_CAM_ENGINE</code>	<code>provider:model</code> for the CAM workbench
<code>WORKFLOWS_DOC_EXPIRY_WARN_DAYS</code> , <code>_DOC_EXPIRY_INTERVAL_HOURS</code>	Document expiry monitor
<code>WORKFLOWS_COVENANT_HORIZON_DAYS</code> , <code>_COVENANT_INTERVAL_HOURS</code>	Covenant monitor
<code>WORKFLOWS_EWS_ASSIGN_SLA_HOURS</code> , <code>_EWS_INVESTIGATION_SLA_HOURS</code> , <code>_EWS_ESCALATED_REMINDER_HOURS</code>	EWS SLAs
<code>WORKFLOWS_VOX_CONFIRM_AMBIGUOUS_COMPANY</code> , <code>_VOX_CONFIRM_LEAD_SELECTION</code>	Whether VocX parks for human confirmation
<code>WORKFLOWS_QUALIFICATION_CHECKLIST</code>	The qualification bar
<code>WORKFLOWS_SEARCH_ATTRIBUTES_ENABLED</code>	Reflect business status into Temporal search attributes
<code>WORKFLOWS_WORKER_BUILD_ID</code> , <code>_METRICS_BIND_ADDRESS</code>	Worker versioning and Prometheus

Other

Variable	Meaning
<code>LMS_ENABLED</code>	false today. The book ends at <code>Disbursed</code> ; no loan account is opened
<code>ADVAYA_URL</code>	The Advaya endpoint (simulated by default)
<code>ATLAS_STAGE_AMBER_DAYS</code> / <code>_RED_DAYS</code>	Ageing thresholds on the dashboard
<code>PGBACKUP_KEEP</code>	Dump retention days (default 14)

3. Fixed configuration worth knowing

These are set in the compose file, not `.env`, but you will need them.

Setting	Value	Note
<code>REGISTER_SERVICE_API_KEYS</code>	<code>svc_atlas:... , svc_vox:... , svc_pulse:... , svc_workflows:... , svc_gateway:... , svc_advaya:...</code>	The named-principal map
<code>REGISTER_STREAM_THROUGH_API</code>	<code>true</code>	Serve document bytes through the API rather than redirecting to a presigned MinIO URL. Without it, "View" 302s to <code>http://minio:9000/...</code> — a docker-internal host the browser cannot resolve — and the download dies silently. It also keeps the one-door posture
<code>REGISTER_WEB_CONCURRENCY</code>	<code>4</code>	
<code>REGISTER_DB_POOL_SIZE</code>	<code>20</code>	Postgres <code>max_connections</code> is capped so <code>pool_size × replicas</code> stays under the ceiling
<code>GATEWAY_CACHE_TTL_S</code>	<code>30</code>	Permission cache
<code>ATLAS_PERMISSION_CACHE_TTL_S</code>	<code>30</code>	
<code>ATLAS_MAX_PAGES_PER_RESOURCE</code>	<code>10</code>	×200 rows per vertical per request
<code>REGISTER_ADVAYA_INTEGRATION_ENABLED</code>	<code>true</code>	The acknowledgement lane, against a simulated peer

4. The production posture — what flips

`docker-compose.prod-posture.yml`, applied as a second `-f`:

Setting	Dev	Production
GATEWAY_REQUIRE_AUTH	false	true
GATEWAY_OIDC_ISSUER	empty	http://dex:5556/dex
GATEWAY_OIDC_ALLOWED_DOMAINS	empty	evamfinance.com
REGISTER_ENFORCE_RBAC	false	true
REGISTER_ENFORCE_RLS	false	true
REGISTER_ONLINE_REVALIDATION	false	true
ACCESS_AUTO_SEED	true	false
WORKFLOWS_REQUIRE_AUTH	false	true
VOCX_DEV_UI	true	false

```
docker compose -f deploy/compose/docker-compose.yml \
    -f deploy/compose/docker-compose.prod-posture.yml \
    --profile sso --profile backup up -d --build
```

--profile sso is required. Dex is profile-gated in the base file and an override cannot un-gate it — Compose filters profiled services out *before* merging. Without the flag you get `REQUIRE_AUTH` on with no issuer reachable, and every request 401s.

5. `STT_CPU_THREADS`, once and clearly

```
# services/stt/app/config.py
cpu_threads: int = 0
```

Value	Behaviour
0	CTranslate2 chooses, based on the host's core count — which it reads regardless of the container's CPU quota
N	Exactly N decoder threads

It changes scheduling only. Accuracy, model, beam width and transcript are untouched.

Deployment	stt CPU limit	Correct value
Helm	limits.cpu: 2	2
Compose	none	Derive from <code>nproc</code> . 0 is often right; 2 on a 4-core box is a throttle

The failure mode 0 guards against: a container limited to 2 cores spawning a thread per *host* core, then having those threads fight over the 2 it actually has — decoding **slower** than 2 threads would.

6. UI build-time variables

The UI is a static bundle, so these are **baked in at build time** — changing them means rebuilding the `ui` image.

Variable	Default	Meaning
<code>VITE_USE_REAL_API</code>	—	<code>true</code> for a real backend
<code>VITE_API_BASE_URL</code>	<code>/v1</code>	The Register prefix
<code>VITE_ACCESS_URL</code>	<code>/access</code>	
<code>VITE_VOCX_URL</code>	<code>/vocx</code>	Rooted at the origin, not under <code>/v1</code>
<code>VITE_VOCX_MAX_SECONDS</code>	<code>180</code>	Recording cap, clamped 30–600

```
# Rebuild after changing any of them
docker compose -f deploy/compose/docker-compose.yml build ui
docker compose -f deploy/compose/docker-compose.yml up -d ui
docker compose -f deploy/compose/docker-compose.yml exec nginx nginx -s reload
# then hard-refresh the browser (Ctrl+Shift+R)
```

Declared as `ARG` / `ENV` pairs in `deploy/ui-image/Dockerfile`.

7. Timeouts — the four places that must agree

Path	Browser	nginx	gateway	Helm
default	axios default	65 s	60 s	main Ingress
<code>/vocx/v1/capture</code>	300 s	305 s	600 s	slow Ingress
<code>/orchestrator/v1/cam/</code>	620 s	625 s	600 s	slow Ingress
<code>/pulse/v1/news/sweep</code>	—	625 s	600 s	slow Ingress

Adding a slow endpoint means editing **all four**:

- `deploy/nginx/nginx.conf` — a `location` with its own `proxy_read_timeout`
- `services/gateway/app/main.py` — `_SLOW_PATHS`
- The browser client — `services/atlas/ui/src/api/*.ts`
- `gateway.ingress.slowPaths` in Helm values

Widening one alone does nothing; the shortest hop decides.

8. Helm values

`deploy/helm/prism/values.yaml`, one block per subchart:

```

global:
postgresql:
register:
access:
gateway:
ui:
vocx:
stt:          # config.cpuThreads: 2  ← matches limits.cpu: 2
pulse:
atlas:
minio:
temporal:
workflows:

```

```
helm upgrade --install prism deploy/helm/prism -n prism --create-namespace -f prod-values.yaml
```

Set `gateway.ingress.slowPaths` from a **values file**, never `--set`. Helm's `--set` grammar mis-parses `slowPaths=[]`; a values file behaves correctly.

Secrets in Kubernetes go in `Secret` objects referenced by the values, not inline.

9. Secrets handling

The three paths that live only on disk, **never** in git or a delivery archive:

```

deploy/compose/.env
deploy/vocx-secrets/
deploy/nginx/certs/

```

Verify every archive you produce:

```

unzip -l "$ZIP" | grep -Ei "client_secret|token\.json|vocx-
secrets/*.json|vocx_tokens|compose/.env"
# exit 1 (no output) = clean

```

`prism-deploy.sh` snapshots all three before an upgrade and restores them into the new tree, failing hard if `.env` is missing afterwards.

10. Configuration checklist for a new production deployment

- ☐ Every secret in §2 changed from its dev default
- ☐ `INTERNAL_SIGNING_SECRET` is long and random — it signs identity
- ☐ `WORKFLOWS_PAYLOAD_ENCRYPTION_KEY` set (32 bytes, base64url) so Temporal history is ciphertext
- ☐ `GATEWAY_REQUIRE_AUTH=true` and an issuer configured
- ☐ `GATEWAY_OIDC_ALLOWED_DOMAINS` set — mandatory with a consumer IdP
- ☐ `REGISTER_ENFORCE_RBAC`, `REGISTER_ENFORCE_RLS`, `REGISTER_ONLINE_REVALIDATION` all `true`
- ☐ `ACCESS_AUTO_SEED=false`

- ☐ `VOCX_DEV_UI=false`
- ☐ `VOCX_REDIRECT_URI` is the public HTTPS URL, matching Google Console exactly
- ☐ `VOCX_S3_PUBLIC_ENDPOINT_URL` is browser-reachable
- ☐ `DEV_PORTS_BIND=127.0.0.1` (or the ports closed at the security group)
- ☐ `STT_CPU_THREADS` derived from `nproc` (compose) or set to `2` (Helm)
- ☐ `--profile sso --profile backup`
- ☐ A CA-issued certificate in `deploy/nginx/certs/`
- ☐ SMTP configured, and a test digest actually received
- ☐ `.env` is `chmod 600` and owned by the deploy user